



Penguin
Random
House

DK LONDON

Senior Art Editor Sheila Collins

Editor Vicky Richards

US Editor Kayla Dugger

US Executive Editor Lori Cates Hand

Picture Researcher Nic Dean

Illustrators Brendan McCaffery,

Adam Benton, Peter Bull, Gus Scott

Managing Editor Francesca Baines

Managing Art Editor Philip Letsu

Senior Production Editor Andy Hilliard

Production Controller Sian Cheung

Jacket Designer Surabhi Wadhwa-Gandhi

Jacket Design Development Manager

Sophia MTT

Publisher Andrew Macintyre

Associate Publishing Director Liz Wheeler

Art Director Karen Self

Publishing Director Jonathan Metcalf

DK DELHI

Senior Editor Virien Chopra

Project Art Editor Baibhav Parida

Project Editor Kathakali Banerjee

Art Editor Sifat Fatima

Picture Researcher Nishwan Rasool

Picture Research Manager Taiyaba Khatoon

Managing Editor Kingshuk Ghoshal

Managing Art Editor Govind Mittal

Senior DTP Designer Neeraj Bhatia

DTP Designer Anita Yadav

Pre-Production Manager Balwant Singh

Production Manager Pankaj Sharma

First American Edition, 2021

Published in the United States by DK Publishing
1450 Broadway, Suite 801, New York, NY 10018

Copyright © 2021 Dorling Kindersley Limited
DK, a Division of Penguin Random House LLC
21 22 23 24 25 10 9 8 7 6 5 4 3 2 1
001-310498-Aug-2021

All rights reserved.

Without limiting the rights under the copyright reserved above, no part of this publication may be reproduced, stored in or introduced into a retrieval system, or transmitted, in any form, or by any means (electronic, mechanical, photocopying, recording, or otherwise), without the prior written permission of the copyright owner.

Published in Great Britain by Dorling Kindersley Limited

A catalog record for this book is available
from the Library of Congress.
ISBN 978-0-7440-2890-4

DK books are available at special discounts
when purchased in bulk for sales promotions,
premiums, fund-raising, or educational use.
For details, contact: DK Publishing Special Markets,
1450 Broadway, Suite 801, New York, NY 10018
SpecialSales@dk.com

Printed and bound in China

For the curious

www.dk.com

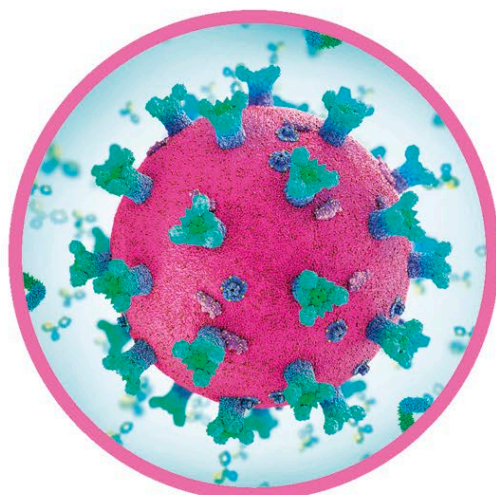


Smithsonian

Established in 1846, the Smithsonian is the world's largest museum and research complex, dedicated to public education, national service, and scholarship in the arts, sciences, and history. It includes 19 museums and galleries and the National Zoological Park. The total number of artifacts, works of art, and specimens in the Smithsonian's collection is estimated at 155.5 million.



This book was made with Forest Stewardship Council™ certified paper—one small step in DK's commitment to a sustainable future. For more information go to www.dk.com/our-green-pledge



PROTECTING AND SURVIVING

120

ANTIBIOTICS	122
VACCINES	126
BLOOD DONATION	128
PACEMAKERS	130
EXOSKELETONS	132
MEDICAL MACHINES	134
COCHLEAR IMPLANTS	136
RUNNING BLADES	138
CRASH TESTING	140
UV RADIATION	142
OXYGEN TANKS	144
OIL SKIMMERS	146
BIOME DOMES	148
BIOROCKS	150
FIRE RETARDANTS	154
CARBON CAPTURE	156

LEARNING AND DISCOVERING

158

UNDERWATER VEHICLES	160
FORECASTING VOLCANIC ERUPTIONS	162
WEATHER PLANES	164
SONAR	166
ICE CORES	168
WEATHER BALLOONS	170
TELESCOPES	172
MICROSCOPES	176
DATING FOSSILS	178
CT SCANS	180
REBUILDING FOSSILS	182
THERMAL IMAGING	184
ROBOT CAMERAS	186
DNA ANALYSIS	190
MRI SCANS	192
ARTIFICIAL SKIN	194
MAPPING MIGRATION	196
CAMERA TRAPS	198
GLOSSARY	200
INDEX	204